Protocol v2.3 — Dynamic Biomarkers of Affective-Cognitive Regulation: A Systematic Map with Quality Assessment

Date: 2026-06-15 **Authors:** Jaba Tqemaladze (GLA), Afaf El Fettahi **Framework:** PRISMA 2020 + SWiM (Synthesis Without Meta-Analysis) + JBI Critical Appraisal **Registration:** OSF (PROSPERO не принимает scoping reviews) **Note:** This review conducts formal critical appraisal (JBI) — exceeding standard PRISMA-ScR requirements. PRISMA 2020 is used for reporting; SWiM guidelines for synthesis without meta-analysis. **Status:** Conditional Accept after Minor Revisions (v2.3)

1. Research Question

> **Which peripheral and central biomarkers index dynamic transitions in affective-cognitive regulation in healthy adults, particularly under acute stress, uncertainty, or cognitive challenge?**

PCC Framework

Component	Definition
Population	Healthy adults (≥18 yr), human studies only
Concept	Dynamic transitions (baseline → activation → adaptation/recovery)
Context	Acute stress, uncertainty, cognitive challenge paradigms + ambulatory real-world assessment

2. Theoretical Framework (Introduction — 1–2 paragraphs only)

The review is grounded in a **dynamic tension framework**:

- **Sensation → Affect:** interoceptive registration → emotional valuation
- **Affect → Cognition:** emotional state → cognitive modulation
- **Cognition → Sensation (feedback):** cognitive reappraisal → interoceptive shift

Each transition is a measurable point where regulatory dynamics can be indexed by biomarkers. The review focuses on **transitions** rather than static baseline levels, because regulation is inherently dynamic.

Note: The broader theoretical architecture (microbiome/macrobiome competition, plant/animal/human hierarchy, bacterial proto-cognition) is reserved for a separate *theoretical manuscript and does not enter the review's inclusion criteria or search strategy.*

3. Inclusion Criteria

#	Criterion
1	**Human studies**, healthy adult participants (≥18 yr)
2	**Mixed clinical-healthy samples** — included ONLY if healthy controls are analysed and reported separately
3	At least one **biomarker measured dynamically** (≥2 time points with a minimum interval of 5 minutes between measurements within a single session OR repeated measures across conditions)
4	At least one **measure of affective or cognitive state** (self-report, behavioural, or physiological)
5	**Dynamic perturbation protocol** — acute stress/challenge/uncertainty paradigm OR ambulatory real-world assessment (not resting state only)
6	**Multi-language** — English, German, French, Chinese, Spanish (machine-assisted translation via DeepL for non-English articles; articles untranslatable after reasonable effort are documented as «language-barrier exclusion»)
7	Peer-reviewed primary research, preprints (medRxiv, PsyArXiv), and doctoral dissertations (ProQuest)

4. Exclusion Criteria

#	Criterion
1	Clinical populations only (psychiatric, neurological, metabolic disorders) — unless healthy controls reported separately
2	Static/baseline biomarker measurement only (single time point)
3	Animal studies
4	Case reports, commentaries, editorials
5	No affective/cognitive outcome measure
6	Pharmacological intervention studies (unless biomarker response to challenge is the primary outcome)
7	Focus exclusively on chronic stress (not acute transitions)
8	Resting-state only (no dynamic manipulation)

5. Dynamic Perturbation Protocols (Explicit List)

Laboratory Protocols

Category	Examples
Acute stress	TSST (Trier Social Stress Test), SECPT (Socially Evaluated Cold Pressor Test), MIST (Montreal Imaging Stress Task)
Cognitive challenge	n-back, Stroop, mental arithmetic, working memory tasks
Emotional challenge	Affective images (IAPS), film clips, mood induction
Social-evaluative stress	Public speaking, social rejection (Cyberball), peer evaluation
Uncertainty	Probabilistic tasks, ambiguity paradigms, unpredictable threat (NPU task)
Emotion regulation	Reappraisal, suppression, distraction (explicit instruction paradigms)

Ambulatory / Real-World Protocols

Category	Examples
Ecological Momentary Assessment (EMA)	Smartphone-based repeated self-report + concurrent physiological sampling
Wearable-based ambulatory monitoring	Apple Watch, Oura Ring, Empatica E4, Biopac BioNomadix — continuous HRV/EDA + EMA prompts
Daily-life stress diaries	End-of-day cortisol (hair/saliva) + daily affect ratings over ≥3 days

Rationale: In 2025–2026, journals with IF>10 prioritise ambulatory data with high ecological validity over purely laboratory designs.

6. Biomarker Domains (Primary Only)

Note: Exploratory domains (SCFAs, bile acids, gut permeability, GABA/glutamate, gut-brain peptides) are removed from the main protocol per peer review recommendation. They remain mentioned in the Discussion as «Future directions» pending adequate dynamic human data.

Domain	Biomarkers
Autonomic regulation	HRV (RMSSD, SDNN, HF-HRV, LF/HF), DFA α1, vagal tone, PEP, EDA/SCR
Endocrine stress	Salivary cortisol (AUCi, AUCg), salivary alpha-amylase (sAA)
Inflammatory	CRP, IL-6, TNF-α
Metabolic	Tryptophan/kynurenine pathway (K:T ratio), quinolinic acid, picolinic acid
Neurophysiological	EEG frontal alpha asymmetry (FAA), LPP, P300, ERN, frontal theta

7. Temporal Structure Coding

Each included study is coded for one or more of the following temporal structures:

Code	Structure	Description
BA	Baseline → Activation only	Pre-post challenge, no recovery measurement
BAR	Baseline → Activation → Recovery	Full reactivity + recovery curve
RM	Repeated measures across conditions	Multiple conditions, each with biomarker sampling
LD	Longitudinal / daily-life dynamics	Multi-day sampling, ecological momentary assessment (EMA)
ESM	Ecological Sampling Method	Continuous wearable data + EMA prompts in real-world settings

ESM code — minimum criteria (must satisfy ALL four): 1. **Duration:** Minimum **3 days** of data collection in the natural environment (not laboratory) 2. **Sampling density:** Minimum **3 measurements per day** (EMA prompts) OR continuous wearable recording with ≥3 analysis epochs per day 3. **Temporal coupling:** Biomarker and affective/cognitive measure are **timestamp-matched** (same hour or synchronised via device clock) 4. **Device specification:** If wearable — model and sampling rate documented (e.g., «Empatica E4, EDA at 4 Hz, HR at 1 Hz»)

This coding serves as a quality/richness indicator and allows subgroup comparison. ESM-coded studies are analysed separately as the highest tier of ecological validity. Studies with <3 days, <3 measurements/day, or uncoupled biomarker-affect sampling are coded as LD (Longitudinal), not ESM.

8. Databases to Search

Database	Access	Priority	Rationale
PubMed/MEDLINE	Full (Jaba)	Primary	Biomedical core
Embase	Request via university/ResearchGate	**Mandatory**	Pharmacological + physiological data; broader than PubMed for biomarker studies
Scopus	Via Afaf's institution	Supplementary	Social + behavioural sciences
PsycINFO	Via Afaf's institution	Supplementary	Psychological constructs, emotion regulation
Web of Science	Limited (Jaba)	Supplementary	Citation tracking
ProQuest Dissertations	Request access	Supplementary	Grey literature — doctoral theses
Google Scholar	Free	Supplementary	First 200 results for grey literature + citation tracking
Backward citation	From included studies + relevant reviews	Supplementary	Snowballing

Rationale: Embase is mandatory for IF>10 journals — it indexes ~2,900 unique journals not in PubMed, with superior coverage of pharmacology, physiology, and drug effects on biomarkers. ProQuest captures unpublished dynamic biomarker studies. Google Scholar (first 200) captures preprints and conference proceedings.

9. Search String (PubMed — v2.2, IF 18+ upgrade)

Restructured per reviewer recommendation: no generic "biomarker" without context. Emotional/cognitive regulation terms moved to the front. MeSH terms for all primary domains.*

```
(("Heart Rate"[MeSH] OR "heart rate variability"[MeSH] OR "HRV"[tiab] OR "RMSSD"[tiab]
   OR "Hydrocortisone"[MeSH] OR "cortisol"[tiab] OR "salivary alpha-amylase"[tiab] OR "sAA"[tiab]
```

OR "Interleukin-6"[MeSH] OR "IL-6"[tiab] OR "C-Reactive Protein"[MeSH] OR "CRP"[tiab] OR "TNF-alpha"[tiab]
OR "Tryptophan"[MeSH] OR "tryptophan"[tiab] OR "Kynurenine"[MeSH] OR "kynurenine"[tiab]
OR "Electroencephalography"[MeSH] OR "EEG"[tiab] OR "frontal alpha asymmetry"[tiab]
OR "LPP"[tiab] OR "P300"[tiab] OR "ERN"[tiab] OR "frontal theta"[tiab]
OR "Galvanic Skin Response"[MeSH] OR "EDA"[tiab] OR "skin conductance"[tiab]))

AND

(("Emotional Regulation"[MeSH] OR "Stress, Psychological"[MeSH] OR "Cognition"[MeSH]
OR "affect* regulation"[tiab] OR "emotion regulation"[tiab] OR "cognitive reappraisal"[tiab]
OR "acute stress"[tiab] OR "social stress"[tiab] OR "uncertainty"[tiab] OR "TSST"[tiab]
OR "cognitive load"[tiab] OR "threat"[tiab] OR "challenge"[tiab]))

AND

(("repeated measures"[tiab] OR "time series"[tiab] OR "recovery"[tiab]
OR "reactivity"[tiab] OR "transition*"[tiab] OR "dynamic*"[tiab]
OR "EMA"[tiab] OR "ecological momentary"[tiab] OR "ambulatory"[tiab]
OR "wearable"[tiab] OR "daily diary"[tiab]))

AND

("Humans"[MeSH] AND "Adult"[MeSH])

AND

("2000"[Date - Publication] : "2026"[Date - Publication])

NOT

("animal*"[ti] OR "review"[pt] OR "meta-analysis"[pt] OR "comment"[pt] OR "editorial"[pt])

Scopus adaptation: Replace MeSH with EMTREE/thesaurus terms. Remove [tiab]/[pt] tags. Use Scopus field codes (TITLE-ABS-KEY, INDEXTERMS).

Embase adaptation: Use EMTREE terms (`'heart rate variability'/exp`, `'hydrocortisone'/exp`, etc.) + `:ti,ab,kw` field qualifiers.

10. Risk of Bias & Quality Assessment (JBI + NIH)

Critical Appraisal Tool (Unified — Single Framework)

Single tool for all studies: JBI Critical Appraisal Checklists (revised 2026) — design-specific modules within a unified methodological framework.

Study design	JBI Module	Items
Cross-sectional (lab challenge, single-session)	JBI Checklist for Analytical Cross-Sectional Studies	8
Quasi-experimental (pre-post challenge, no randomisation)	JBI Checklist for Quasi-Experimental Studies	9
Cohort/longitudinal (multi-day, ESM, ambulatory)	JBI Checklist for Cohort Studies	11

Rationale for unified JBI (not NIH + JBI mix): - JBI revised 2026 provides design-specific modules within a **single methodological framework** — same scoring philosophy across designs - Enables direct quality comparison across lab and ambulatory studies - NIH Tool has not been revised since 2014 and lacks design-specific modules - Using two different tools (JBI + NIH) on different study subsets would prevent cross-design quality comparison

Procedure

1. Both reviewers independently assign each study to a design category and apply the corresponding JBI module
2. Quality score normalised as **% of applicable items** (e.g., 7/8 = 87.5%) for cross-design comparability
3. Inter-rater agreement for quality ratings reported (Cohen's κ or weighted κ)
4. **Quality score does NOT exclude studies** — it informs the narrative synthesis (e.g., «high-quality studies (JBI ≥ 80%) consistently show X, while lower-quality studies show mixed results»)
5. Quality-by-domain heatmap presented in supplementary materials

11. Screening Procedure

Step	Action	Who
0	**OSF Registration** — register protocol BEFORE any screening	Jaba

Step	Action	Who
	begins. Deadline: 2026-06-22.	
1a	**Pilot calibration** — both reviewers independently screen 50 random titles/abstracts. Calculate Cohen's κ. If κ < 0.70 → discuss discrepancies, refine criteria, repeat pilot. Proceed to full screening only when κ ≥ 0.70.	Both
1	**Deduplication**	Rayyan auto + manual
2	**Title/abstract screening**	Both independently (Rayyan)
3	**Conflict resolution**	Discussion; if unresolved → documented as discrepancy
4	**Full-text retrieval** (including DeepL-assisted translation for non-English articles)	Both reviewers
5	**Full-text screening**	Both independently
6	**Quality assessment** (JBI/NIH — both independently, see §10)	Both
7	**Final inclusion**	Consensus
8	**PRISMA 2020 flow diagram**	Standard format (includes database + grey literature + register branches)

Inter-rater agreement: Cohen's κ reported for title/abstract, full-text, and quality assessment stages. Pilot calibration κ also reported.

Conflict resolution: Discrepancies resolved by discussion. If no consensus → study flagged as «uncertain» and reported transparently.

12. Extraction Table

Field	Description
Study ID	First author, year, DOI

Field	Description
Sample	N, age (M±SD), sex (%F), population type
Design	Paradigm, conditions, time points
Perturbation type	Stress / cognitive / emotional / social / uncertainty / emotion regulation / ambulatory-EMA
Perturbation duration	Minutes (lab) or days (ambulatory)
Setting	Laboratory / Ambulatory / Hybrid
Biomarker(s)	Type, sample matrix, measurement method, sampling times
Temporal structure	BA / BAR / RM / LD / ESM (see §7)
Affective/cognitive measure	Instrument, timing
Key finding	Direction, effect size (if available), statistical test
Dynamic feature	Transition type (baseline→peak, peak→recovery, etc.)
Quality score	JBI / NIH score (§10)
Language	Original language; translation method if non-English

13. Synthesis Plan

1. **Narrative synthesis** — structured by biomarker domain (§6)
2. **Transition matrix** — biomarker domain × temporal structure (BA / BAR / RM / LD / ESM)
3. **Setting comparison** — Laboratory vs. Ambulatory (ESM) — which biomarkers replicate across settings?
4. **Quality-stratified synthesis** — high-quality (JBI ≥ 7/8) vs. lower-quality studies compared within each domain
5. **Gap analysis** — which transitions, biomarker domains, and settings are understudied?
6. **Feasibility assessment** — is a future meta-analysis of reactivity slopes viable for specific biomarker–challenge combinations?
7. **Future directions** — gut-derived biomarkers (SCFAs, bile acids, zonulin, GABA/glutamate) as emerging candidates pending adequate dynamic human data

Unit-of-analysis rule: If a study reports ≥3 biomarkers from different domains, each biomarker-domain pair is coded separately. Primary synthesis uses one biomarker per study per domain (pre-specified hierarchy: HRV > cortisol > IL-6 > CRP > EEG > tryptophan/kynurenine).

14. Target Journals (Re-ranked for IF 18+ Aspirational Trajectory)

Tier	Journal	IF	Rationale
T1 — aspirational	*Neuroscience & Biobehavioral Reviews*	8.5	Premier review journal; requires quality assessment + multi-database search
T1 — aspirational	*Molecular Psychiatry*	11.0	Biomarker focus; requires genetic/molecular depth
T2 — realistic	*Psychoneuroendocrinology*	3.5	Best thematic fit: stress + biomarkers + dynamic
T2 — realistic	*Biological Psychiatry: CNNI*	5.5	Cognitive neuroscience + neuroimaging + biomarkers
T2 — realistic	*Psychophysiology*	3.7	HRV/EEG/ERP methodology focus
T3 — safety	*Biological Psychology*	3.0	Affective-cognitive regulation + peripheral physiology
T3 — safety	*Comprehensive Psychoneuroendocrinology*	new	Open access, scoping-review friendly

Final choice after feasibility phase — once volume and type of eligible studies is known. T1 submission only if ≥80 eligible studies with ≥30% ESM/ambulatory designs and adequate quality (JBI ≥ 6/8 in ≥50% of studies).

15. Feasibility Phase (Before Full Commitment) — Enhanced

Step	Action	Who	Target
1	**Protocol v2.2 approved**	Afaf reviews	This week
2	**Pilot PubMed search** (v2.2 string)	Jaba	~3 days
3	**Pilot Embase search**	Request access; run if granted	~2 weeks
4	**Pilot Scopus search**	Afaf	~1 week
5	**Estimate eligible studies** from pilot (title-only screen)	Both	After searches

Step	Action	Who	Target
6	**Pilot report** — N hits, N after title screen, estimated N eligible, language distribution, setting distribution (lab vs ambulatory)	Jaba drafts	After step 5
7	**Decision**	Both	Scoping review GO / narrow further / pause

Decision criteria (tightened per IF 18+ reviewer): - **GO — T1 target** if ≥80 eligible studies with ≥30% ESM/ambulatory designs, and ≥50% studies expected to score JBI ≥ 6/8 - **GO — T2 target** if ≥50 eligible studies with adequate diversity across biomarker domains - **NARROW** if >200 hits — focus on 2 primary domains (HRV + cortisol) - **PAUSE** if <30 eligible studies — insufficient literature; consider expanding to clinical populations

The pilot report is submitted with the protocol to the target journal as evidence of feasibility (reviewer recommendation).

16. Roles (Preliminary)

Role	Who
Theoretical framing: emotion-cognition loop, regulatory dynamics	Afaf
Microbiome-metabolite side, HRV methodology, BioSense pipelines	Jaba
Protocol drafting	Jaba drafts, Afaf edits
Database search execution	Jaba (PubMed, Embase, Google Scholar), Afaf (Scopus, PsycINFO, ProQuest)
Screening (dual independent)	Both
Quality assessment (JBI/NIH — dual independent)	Both
Synthesis + manuscript	Joint
Authorship	Joint first authorship (order TBD)

17. Timeline (Flexible — All Negotiable)

Step	Target
Afaf feedback on protocol v2.2	This week
Revised protocol finalised	~1 week
OSF registration	After protocol finalised
Pilot searches (PubMed, Scopus, Embase)	Weeks 1–3
Pilot report	Week 3
Rayyan project created + references uploaded	Afaf, after pilot report
Title/abstract screening	Weeks 4–6
Full-text screening + quality assessment (JBI/NIH)	Weeks 7–10
Extraction	Weeks 10–12
Synthesis + manuscript	Weeks 13–16

Data Management Plan

- All screening decisions stored in Rayyan (cloud backup)
- Extraction data stored in Google Sheets (shared, version-controlled)
- Quality assessment ratings stored in Google Sheets with dual-entry verification
- Protocol and amendments versioned on OSF
- Final dataset published as supplementary material or on OSF
- Non-English articles: DeepL-assisted translation; original + translated excerpts stored

Appendix: Change Log

v2.2 → v2.3 (second peer review: Minor Revisions, 2026-06-15)

#	Change	Rationale
1	**Unified JBI (single tool)** — JBI revised 2026 for ALL studies (cross-sectional + quasi-experimental + cohort modules). Removed NIH Tool. Quality normalised as %.	Cross-design quality comparison requires a single methodological framework; NIH not revised since 2014

#	Change	Rationale
2	**ESM operationalised** — 4 minimum criteria: ≥3 days, ≥3 measurements/day, timestamp-matched, device specification	Prevents dilution of ESM construct by weak designs (e.g., 2 time points across 2 days)
3	**PRISMA 2020 + SWiM** — upgraded from PRISMA-ScR. Formal JBI critical appraisal exceeds scoping review standard.	Accurate reporting framework for systematic map with quality assessment

v2.1 → v2.2 (peer review: IF 18+ Major Revisions, 2026-06-15)

#	Change	Rationale
1	**Removed «English only» →** multi-language (EN, DE, FR, ZH, ES) with DeepL	Publication bias; IF 18+ requirement
2	**Added Embase** as mandatory database	~2,900 unique journals not in PubMed; required for biomarker studies
3	**Added ProQuest Dissertations + Google Scholar** (first 200)	Grey literature capture; reduces file-drawer bias
4	**Replaced search string** with reviewer-provided MeSH-centric version	Removed generic «biomarker*»; added emotional regulation MeSH; added EMA/wearable terms
5	**Renamed «Challenge Paradigms» → «Dynamic Perturbation Protocols»**	Terminology alignment with IF 10+ literature
6	**Added Ambulatory/Real-World section** (EMA, wearables, daily diaries)	Ecological validity; IF 18+ journals prioritise ambulatory data
7	**Added «ESM» code** to Temporal Structure coding	Ecological Sampling Method — highest validity tier
8	**Removed Exploratory biomarker domains** from main protocol → Discussion «Future directions»	Per reviewer: «including domains without adequate data reduces credibility»
9	**Upgraded Quality**	Formal Risk of Bias; mandatory for IF

#	Change	Rationale
	Assessment → dual JBI + NIH tools with dual independent rating	10+
10	**Added pilot report requirement** → submit with protocol as feasibility evidence	Reviewer: «show that N≈50, not N≈5000»
11	**Tightened Go/No-Go criteria** → T1/T2 tiers with ESM and quality thresholds	Aspirational tiering for IF 18+ trajectory
12	**Added Embase/Scopus search string adaptation notes**	Reproducibility across databases
13	**Extraction table:** added Setting, Quality score, Language fields	Traceability for multi-language and ambulatory data

v2.0 → v2.1 (MBPR peer review: 78.7/100, 2026-06-15)

#	Change
1	Added MeSH terms to PubMed search string for all primary domains
2	Added pilot calibration step (50 abstracts, κ ≥ 0.70)
3	Inclusion criterion #3: added minimum 5-min interval between time points
4	Added date filter 2000–2026 to search string
5	Added OSF registration deadline (2026-06-22) as Step 0
6	Added unit-of-analysis rule for multi-biomarker studies
7	Added JBI Critical Appraisal for primary domains
8	Added Data Management Plan section

v1 → v2 (Afaf’s 8 methodological comments, 2026-06-15)

#	Change
1	PROSPERO → OSF registration
2	Mixed samples: only if healthy controls reported separately
3	Temporal structure: added 4-category coding (BA / BAR /

#	Change
	RM / LD)
4	Challenge paradigms: explicit list with examples
5	Biomarker primary vs exploratory: split into two tiers
6	Search string: added paradigm terms (TSST, social stress, etc.)
7	Third reviewer not blocking: documented dual screening + conflict resolution
8	Target journals: expanded list with ranking
9	Feasibility phase: added explicit Go/No-Go criteria